



Hack<IT>

Root Cause Analysis



12 September 2026



NB! Please turn on your captions/
Live captions for those who are
unable to hear the presenter



Hack<IT>

Root Cause Analysis



AGENDA



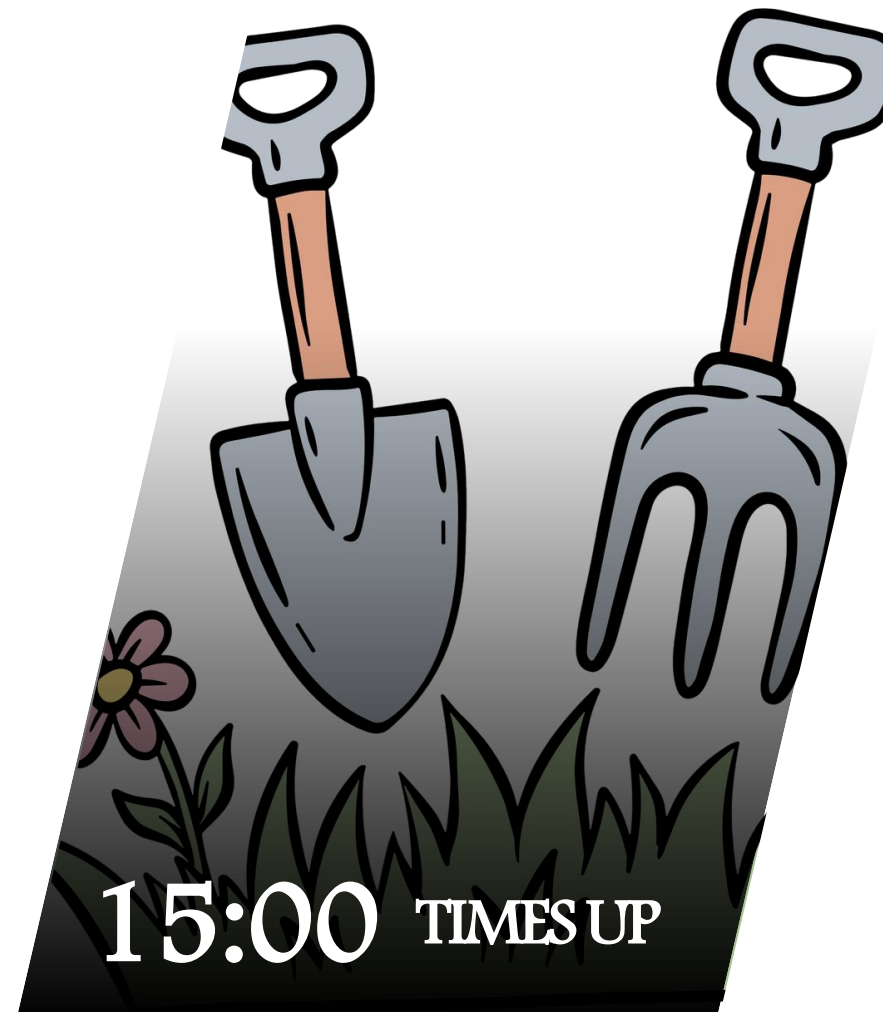
10:00 KICK-OFF

Hack<IT> kick-off! Introduction and problem statement.

We enter the final hour, and the leaderboard is frozen.



14:00 FINAL HOUR



15:00 TIMES UP

The hackathon time is up! Fingers off keyboards.

The top 5 are announced



15:15 TOP 5 ANNOUNCEMENT

OUR PROFESSIONAL TEAM



BIANCA



JESSICA



HUSNAA



KINESH



UHONE

OUR PROFESSIONAL TEAM



DOUGLAS



FRANCOIS



DARSHAN



LIJANI



PIERRE

WELCOME TO HACK<IT>

- Standalone Optimisation Problem
- Anyone can participate
- Solo Participation (No Team Entries)

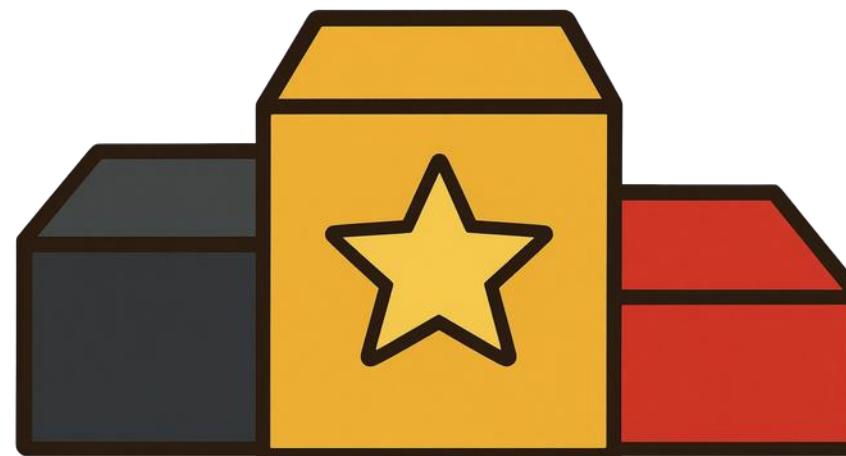


TOP 5

Each Top 5 gets a 4~day pass to Comic Con Africa JHB, from 24th ~
27th September



Winners and prizes will be announced at Comic Con Africa JHB on
Saturday 26th September on the Main stage at 11:30am



WINNER PRIZES



1st Place
R35 000



3rd Place
R8 500



2nd Place
R17 500



4th Place
R5 500



5th Place
R3 500



INTRODUCTION

You are an ecologist working on the planet Photospheria, and your job is to build a biological sample from the planet's plant life and bring it back to Earth for study.



THE PROBLEM



The plant life on Photospheria works through interaction. Different plants affect each other depending on where they are placed. Certain combinations allow new species to flourish.



Other combinations make it more difficult for plants to flourish. Because of this, what you choose to place together directly shapes which kinds of plants exist in your final sample.

THE MISSION

Your main goal is to create the most useful possible sample for Earth's scientists.

There is also a secondary goal: longevity.

Your task is simple:

- Build the best possible sample of Photospherian plant life.
- Make it as diverse as possible.
- See if you can keep as many species alive for as long as possible.



PROBLEM MECHANICS

Grid Layout:

The world is represented by a coordinate system

Ticks (Days):

The simulation progresses in discrete time steps called ticks. Tick 1 represents Day 1.

Tick T represents the final day. Participants may schedule planting actions on any tick from 0 (initial setup) to $T-1$, final actions occur on the penultimate day, and scoring is done on the state of the world on the final day.

A planting action specifies:

- The tick at which to place.
- The plant index (matching the index value in the supported plant catalogue).
- The X coordinate.
- The Y coordinate.

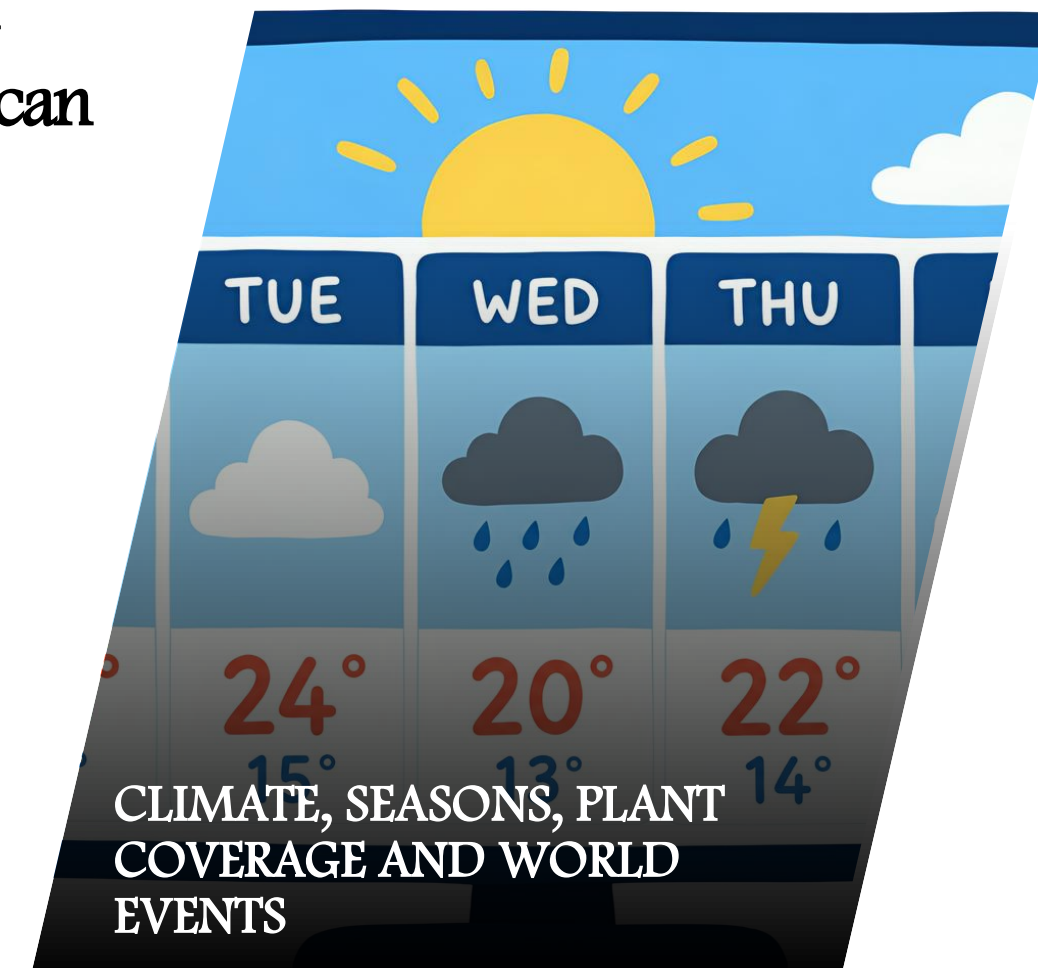
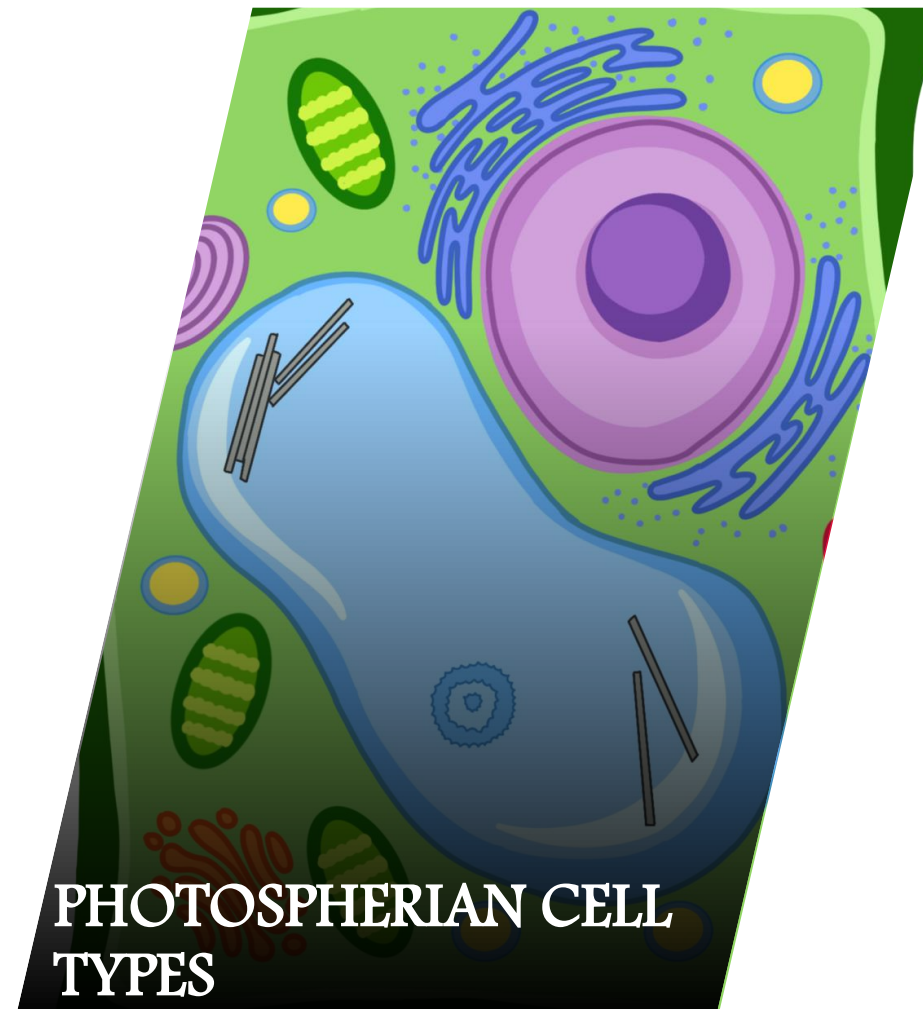


CONSTRAINTS

These constitute the cells in which you will be allowed to place plants. Each plant also has preferred soil types it can spread into.



Each plant species is defined by a set of basic properties that determine how it grows, spreads, and competes within the ecosystem.



The world of Photospheria is subject to various environmental conditions and weather events that affect plant growth, spread, and survival.

CONSTRAINTS

Most plants available in the game cannot be planted unless certain conditions are met.



There is an abundance of wildlife that also has a significant effect on how the plants grow and spread.



There are many different terrain types. Plants can however only live on soil.

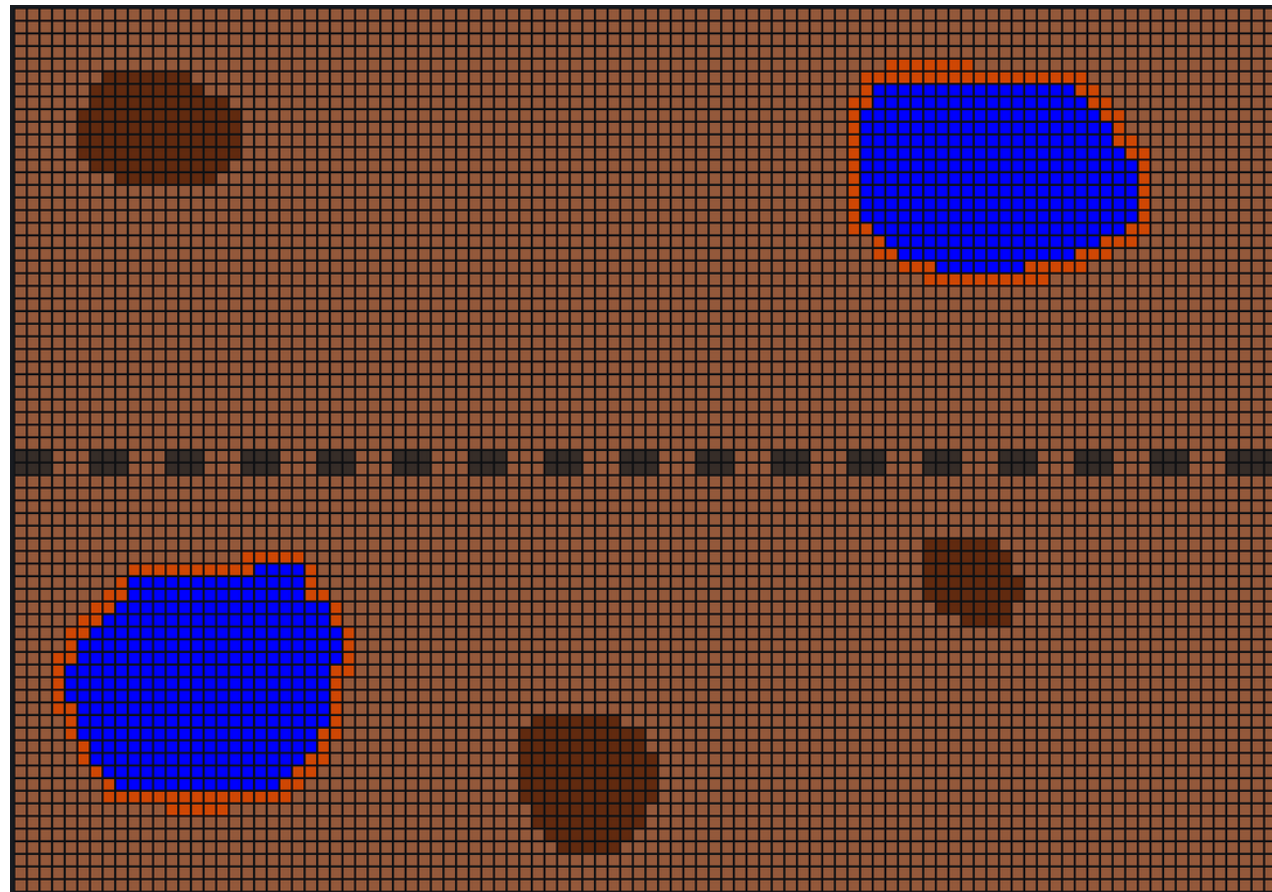
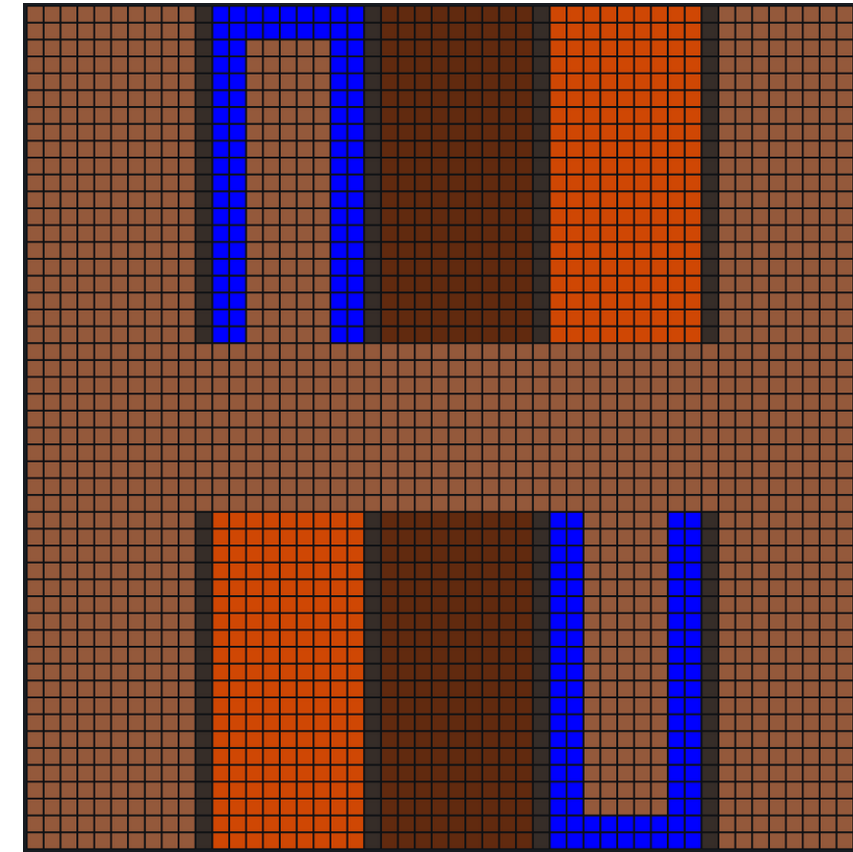


THE LEVELS



Level 1: Greenhouse Study

You will grow your plants in a greenhouse to guard against animals. You will only work with a small subset of plants and your aim is to grow a diverse garden.



Level 2: Garden Growth Study

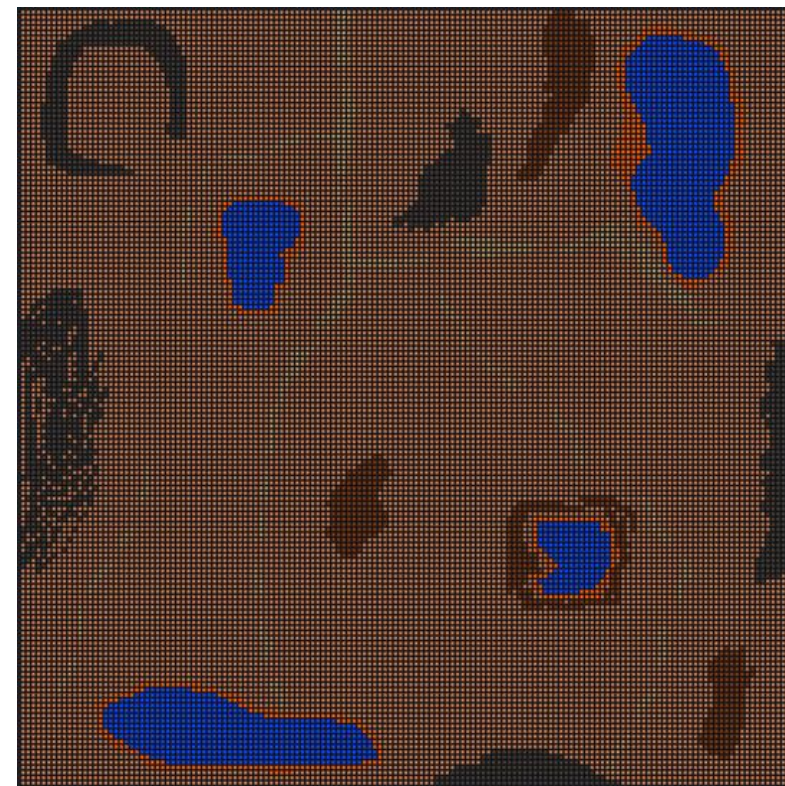
Now that you have the hang of things, we'll move outside into a small garden. Your plants are now exposed to the elements, and a few critters might come and say hi. To add to this, you also have a wider variety of plants available because you have more space.



THE LEVELS

Level 3: Park Potential

Let's kick it up a notch and take it out to a park where we introduce plant unfriendly terrain and more weather events will start to take place.



Level 4: Forest

You are exposed to all the elements in Photospheria, and will be allowed to make use of an entire forest. Use it wisely!



SUBMISSIONS

Participants must submit their solution on the Entelect Hackathon website. The submission must include two items:

- A ZIP of the source code.
- A .json file containing the output.
- The solution must be deterministic, meaning that given the same input, the program must always produce the same output. During validation, the submitted source code will be executed to reproduce the results provided in the submission. If the source code does not produce the same output as the submitted file, the submission will be considered invalid.

code.zip



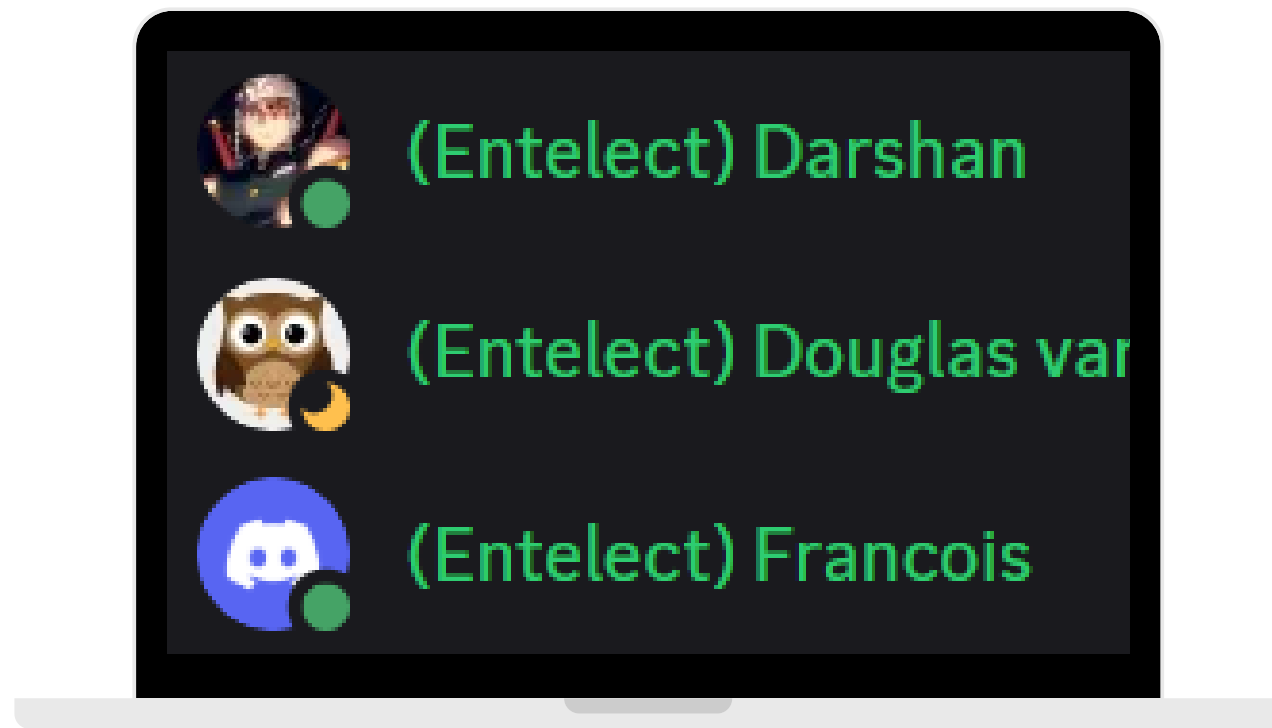
solution.json



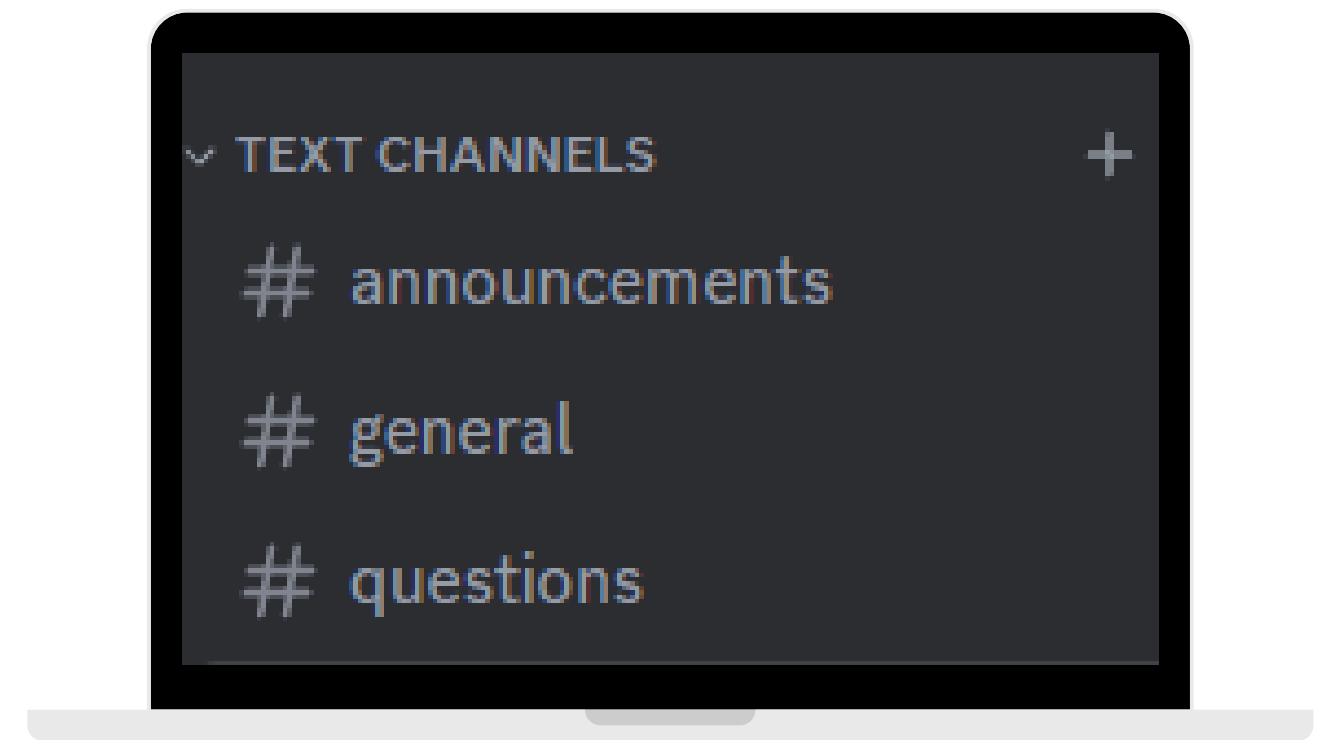
CONTACT US



Discord direct messages



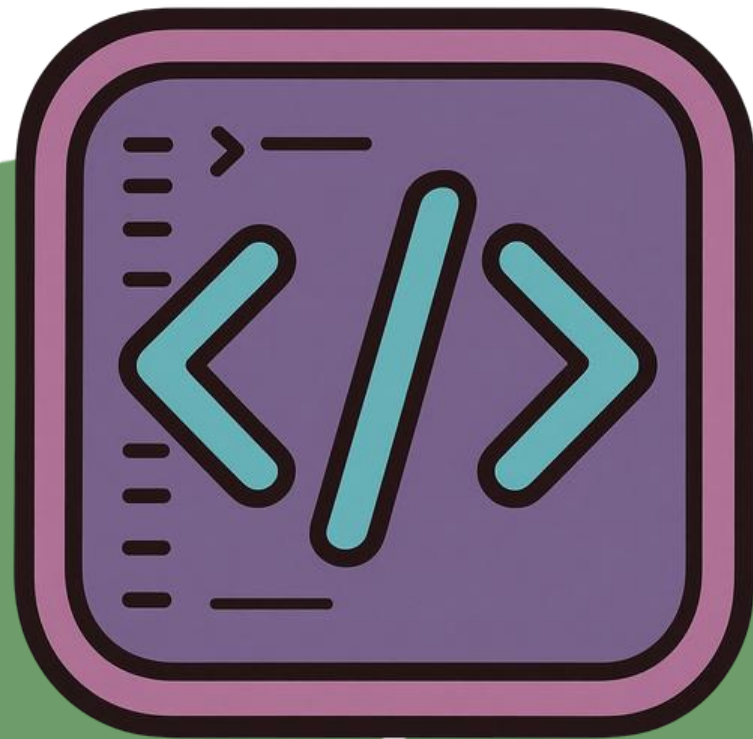
Discord text channels



IMPORTANT!

We will verify the source code with the solution provided. Please ensure that they match and that the code produces the same output.

It needs to be deterministic.



GOOD
LUCK!

